

A convergent series for the Stirling transform of the double factorials:

a proof of Rattaggi's conjecture on OEIS A305404

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Abstract

OEIS A305404 has exponential generating function $1/\sqrt{3-2e^x}$. In October 2020 Diego Rattaggi conjectured that its terms are given by the convergent series $a(n) = \sum_{k \geq 0} k^n \binom{2k}{k} / (2^k 3^{k+1/2})$. We prove it. The argument does not manipulate the sequence at all: instead we form the exponential generating function of the right-hand side and exchange the order of summation. The inner sum is then the central binomial series $\sum_k \binom{2k}{k} z^k = (1-4z)^{-1/2}$ evaluated at $z = e^x/6$, and since $1 - 4 \cdot \frac{e^x}{6} = \frac{3-2e^x}{3}$ the factor $\sqrt{3}$ cancels exactly, returning the entry's own generating function. The exchange is justified by absolute convergence on $|x| < \log(3/2)$, which is precisely the disc on which the generating function is analytic, so comparing Taylor coefficients proves the identity for every n at once.

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1 The sequence and the conjecture

OEIS A305404 is the Stirling transform of the double factorials $(2k-1)!!$; equivalently $a(n) = \sum_{k=0}^n S(n, k) (2k-1)!!$, where $S(n, k)$ are the Stirling numbers of the second kind. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, 1, 4, 25, 217, 2416, 32839, 527185, \dots$$

The entry records the exponential generating function

$$A(x) = \sum_{n \geq 0} a(n) \frac{x^n}{n!} = \frac{1}{\sqrt{3-2e^x}}, \quad (1)$$

and carries the following comment:

Conjecture: $a(n) = \text{Sum}_{k \geq 0} k^n * \text{binomial}(2k, k) / (2^k * 3^{(k+1/2)})$. — *Diego Rattaggi*, Oct 11 2020

As of the “Last modified” line on the live entry (16 February 2025, revision 29) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Write

$$S(n) := \sum_{k \geq 0} \frac{k^n \binom{2k}{k}}{2^k 3^{k+1/2}} = \frac{1}{\sqrt{3}} \sum_{k \geq 0} \binom{2k}{k} \frac{k^n}{6^k}. \quad (2)$$

Since $\binom{2k}{k} \leq 4^k$, the series converges for every n : its terms are $O(k^n (2/3)^k)$. The claim to be proved is $S(n) = a(n)$ for all $n \geq 0$.

2 The interchange

The point of the proof is that k^n is exactly what an exponential generating function produces from a geometric-type series. Nothing about the sequence itself is used.

Lemma 1. For $|z| < \frac{1}{4}$, $\sum_{k \geq 0} \binom{2k}{k} z^k = \frac{1}{\sqrt{1-4z}}$, the series converging absolutely.

Proof. This is the binomial series for $(1-4z)^{-1/2}$: the coefficient of z^k in $(1-4z)^{-1/2}$ is $\binom{-1/2}{k}(-4)^k = \binom{2k}{k}$. Since $\binom{2k}{k} \leq 4^k$, the radius of convergence is $\frac{1}{4}$ and convergence is absolute inside it. $\square$

Lemma 2. For every real x with $|x| < \log \frac{3}{2}$,

$$\sum_{n \geq 0} S(n) \frac{x^n}{n!} = \frac{1}{\sqrt{3}} \sum_{k \geq 0} \binom{2k}{k} \frac{e^{kx}}{6^k},$$

and both sides converge absolutely.

Proof. Substituting (2) and expanding, the double series is

$$\frac{1}{\sqrt{3}} \sum_{n \geq 0} \sum_{k \geq 0} \binom{2k}{k} \frac{1}{6^k} \cdot \frac{(kx)^n}{n!}.$$

All terms are non-negative once x is replaced by $|x|$, so by Tonelli's theorem for double series the order may be exchanged provided the sum of absolute values is finite. Summing over n first gives $\sum_n (k|x|)^n/n! = e^{k|x|}$, so the total is

$$\frac{1}{\sqrt{3}} \sum_{k \geq 0} \binom{2k}{k} \left(\frac{e^{|x|}}{6}\right)^k,$$

which by Lemma 1 is finite exactly when $e^{|x|}/6 < \frac{1}{4}$, that is when $|x| < \log \frac{3}{2}$. Under that condition the interchange is legitimate and summing over n first yields the stated right-hand side. $\square$

3 The theorem

Theorem 3. For every $n \geq 0$,

$$a(n) = \sum_{k \geq 0} \frac{k^n \binom{2k}{k}}{2^k 3^{k+1/2}}.$$

That is, Rattaggi's conjecture on OEIS A305404 is true.

Proof. Fix x with $|x| < \log \frac{3}{2}$. By Lemma 2 and then Lemma 1 applied at $z = e^x/6$ — which satisfies $|z| < \frac{1}{4}$ precisely because $|x| < \log \frac{3}{2}$ —

$$\sum_{n \geq 0} S(n) \frac{x^n}{n!} = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{1-4 \frac{e^x}{6}}}.$$

Now compute the radicand:

$$1 - \frac{4e^x}{6} = 1 - \frac{2e^x}{3} = \frac{3-2e^x}{3},$$

so that

$$\frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{(3-2e^x)/3}} = \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3-2e^x}} = \frac{1}{\sqrt{3-2e^x}} = A(x)$$

by (1). The two factors of $\sqrt{3}$ cancel exactly; this is the whole content of the conjecture.

Both $\sum_n S(n)x^n/n!$ and $A(x)$ are therefore analytic on $|x| < \log \frac{3}{2}$ and agree there, so their Taylor coefficients at 0 agree: $S(n)/n! = a(n)/n!$ for every n . $\square$

Remark 4. The disc of convergence is not an artefact. The function $1/\sqrt{3-2e^x}$ has its nearest singularity where $3-2e^x = 0$, i.e. at $x = \log \frac{3}{2}$, which is exactly the radius produced by the interchange. The conjecture is thus an identity between a function and a series with the same natural domain, and the constant $3^{k+1/2}$ in the denominator — which looks arbitrary as posted — is precisely the $\sqrt{3}$ needed to normalise $(1-4z)^{-1/2}$ at $z = e^x/6$.

4 Verification

Every claim above was checked numerically, by a script written separately from this note.

The generating function identity of Theorem 3 was verified symbolically: $1/\sqrt{3-2e^x}$ and $\frac{1}{\sqrt{3}}(1-4e^x/6)^{-1/2}$ simplify to the same expression.

The series itself was then summed independently, in 40-digit floating point and without reference to the generating function, truncating at $k = 4000$ (the tail is $O(k^n(2/3)^k)$ and far below the working precision). For $n = 0, \dots, 6$ it returns

$$1, 1, 4, 25, 217, 2416, 32839,$$

agreeing with the entry's published terms in every case to more than 12 decimal places.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A305404, conjecture of D. Rattaggi, 11 October 2020; sequence A001147.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999, Chapter 5 (exponential generating functions and the Stirling transform).
- [3] W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976, Theorem 8.3 (rearrangement of absolutely convergent double series).